

Flexible Moral Hazard Problems

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THE INFORMATIVENESS PRINCIPAL

Classic moral hazard: effort is binary or an interval.

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This *informativeness principle* is a key tenant of contract theory:

*“In the late 1970s, Bengt Holmström demonstrated how a principal (e.g., a company’s shareholders) should design an optimal contract for an agent (the company’s CEO), whose action is partly unobserved by the principal. Holmström’s *informativeness principle* stated precisely how this contract should link the agent’s pay to performance-relevant information. Using the basic principal-agent model, he showed how the optimal contract carefully weighs risks against incentives. In later work, Holmström generalised these results to more realistic settings...”*

(The Swedish Academy’s 2016 nobel prize press release)

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Innes (1990): Get monotonicity from free money burning.

THIS PAPER: FLEXIBLE OUTPUT CHOICE

We allow agent to choose *any* output distribution.

Main results:

- Contracts motivated by agent's marginal costs.
- FOSD monotone costs $\implies$ increasing contracts.
- Result by omission: Informativeness plays no role.

Two Examples

COMMON SETUP FOR EXAMPLES

A principal (she) contracts with an agent (he).

- Compact set $X \subset \mathbb{R}$ of possible outputs.
- Principal offers agent a (bounded) contract: $w : X \rightarrow \mathbb{R}$.
- Agent can opt out and get u_0 .
- If opts in, agent covertly chooses $\alpha \in \mathcal{A} \subseteq \Delta(X)$.
- Effort costs: $C : \mathcal{A} \rightarrow \mathbb{R}_+$, increasing in FOSD.
- Payoffs:

$$\text{Principal: } x - w \qquad \text{Agent: } u(w) - C(\alpha).$$

u : strictly increasing, differentiable, unbounded, concave.

STANDARD BINARY EFFORT MODEL

$$X = [L, H], \quad \mathcal{A} = \{\alpha_l, \alpha_h\}.$$

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$$\begin{aligned} \min_{w(\cdot)} & \int w(x) \alpha_h(\mathrm{d}x) \\ \text{s.t.} & \int u \circ w(x) (\alpha_h - \alpha_l)(\mathrm{d}x) \geq C(\alpha_h) - C(\alpha_l) & (\text{IC}) \\ & \int u \circ w(x) \alpha_h(\mathrm{d}x) - C(\alpha_h) \geq u_0. & (\text{IR}) \end{aligned}$$

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$$\frac{1}{u'(w(x))} = \lambda + \mu \left[1 - \frac{f_l(x)}{f_h(x)} \right]$$

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So: w is monotone $\iff$ MLRP holds.

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Implications:

- Cost minimization is trivial: $\min w(L)$ s.t. IR.
- Shape of contract determined by C' and u .
- IC contracts are monotone:

$$w(H) = u^{-1}(u \circ w(L) + C'(\alpha)) \geq u^{-1}(u \circ w(L)) = w(L).$$

OUR MAIN MODEL

A principal (she) contracts with an agent (he).

- Compact set $X \subset \mathbb{R}$ of possible outputs.
- Principal offers agent a (bounded) contract: $w : X \rightarrow \mathbb{R}_+$.
- Limited liability: $w(\cdot) \geq 0$.
- Agent covertly chooses $\alpha \in \mathcal{A} = \Delta(X)$.
- Effort costs: $C : \mathcal{A} \rightarrow \mathbb{R}_+$, continuous, increasing in FOSD.
- Payoffs:

$$\text{Principal: } x - w \qquad \text{Agent: } u(w) - C(\alpha),$$

u : increasing, continuous, unbounded & $u(0) = 0$.

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Assumption. (smoothness) C is **Gateaux differentiable**: every α admits a bounded $k_\alpha : X \rightarrow \mathbb{R}$ s.t.

$$\lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} [C(\alpha + \epsilon(\beta - \alpha)) - C(\alpha)] = \int k_\alpha(x) (\beta - \alpha)(dx)$$

for all $\beta \in \mathcal{A}$.

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Notes:

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Notes:

- $k_\alpha(x)$: MC of increasing probability of output x .
- Without loss of generality: Normalize $\int k_\alpha(X) = 0$.

EXAMPLE 1: FINITE OUTPUTS

Suppose $X = \{x_1, \dots, x_N\}$.

Think of $\alpha = (\alpha_1, \dots, \alpha_N)$ as an element of $\mathbb{R}^N$.

Then Gateaux differentiability $\iff$ differentiable.

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In this case: C convex $\implies C$ differentiable Lebesgue a.e.

EXAMPLE 2: AFFINE COSTS

Suppose we have an increasing and bounded function,

$$k : X \rightarrow \mathbb{R}_+,$$

with $k(\min X) = 0$, and costs are given by

$$C(\alpha) = \int k(x)\alpha(dx).$$

This cost is convex and differentiable, with derivative

$$k_\alpha(\cdot) = k(\cdot).$$

EXAMPLE 3: TRANSFORMATIONS OF AFFINE COSTS

Consider k as before, and let

$$\Psi : \mathbb{R} \rightarrow \mathbb{R}$$

be increasing, convex, differentiable, and $\Psi(0) = 0$. Then,

$$C(\alpha) = \Psi \left(\int k(x) \alpha(dx) \right)$$

satisfies our assumptions. The derivative is

$$k_\alpha(\cdot) = \Psi' \left(\int k(x) \alpha(dx) \right) k(\cdot).$$

Cost Minimization

FIRST-ORDER APPROACH

Lemma. For a bounded $v : X \rightarrow \mathbb{R}$, and $\alpha \in \mathcal{A}$,

$$\alpha \in \arg \max_{\beta \in \mathcal{A}} \left[\int v(x) \beta(\mathrm{d}x) - C(\beta) \right]$$

if and only if

$$\alpha \in \arg \max_{\beta \in \mathcal{A}} \left[\int v(x) \beta(\mathrm{d}x) - \int k_{\alpha}(x) \beta(\mathrm{d}x) \right]$$

(the “only if” direction also works if C is not convex)

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Consider the problem:

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Standard way of writing FOC for optimal $a^* \in (0, 1)$ is

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An equivalent way of writing the above condition is:

$$a^* \in \operatorname{argmax}_{a \in [0,1]} [av - ac'(a^*)].$$

The lemma generalizes the second formulation.

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CHARACTERIZATION OF IC

Say a contract-distribution pair (w, α) is **IC** if

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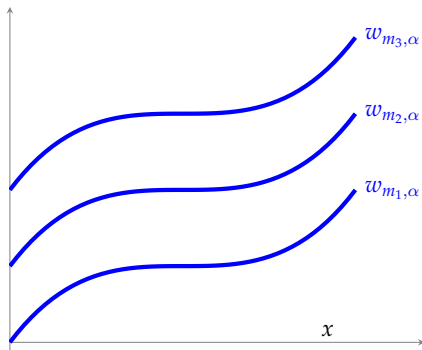
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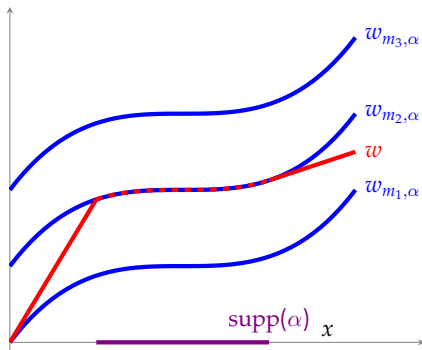
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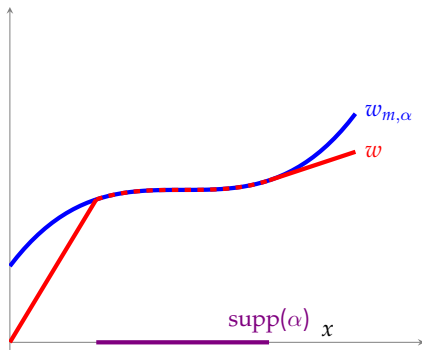
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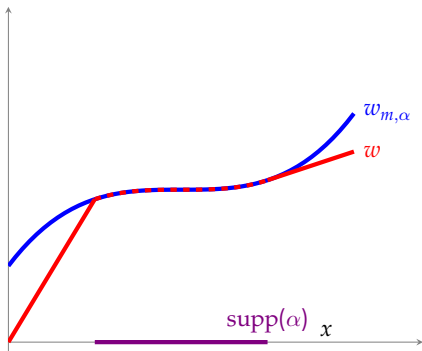


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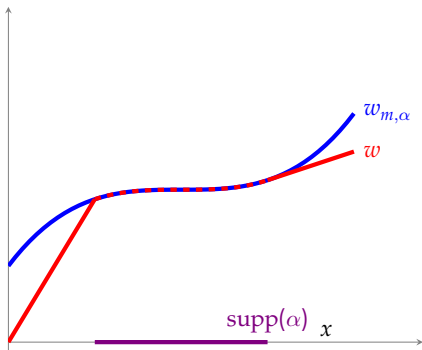
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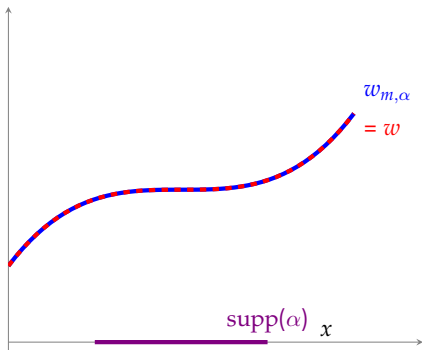
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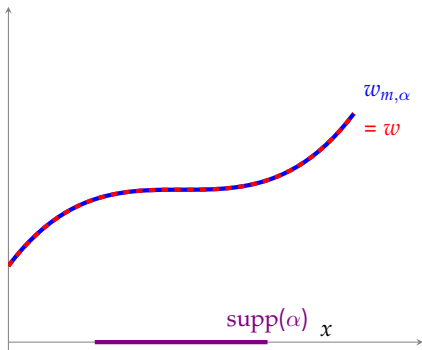
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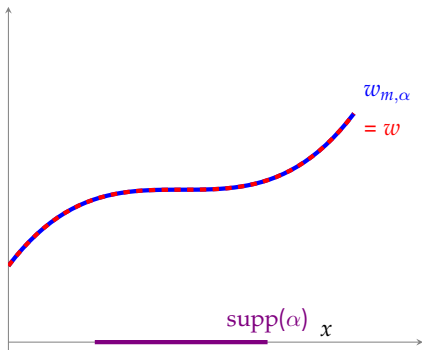
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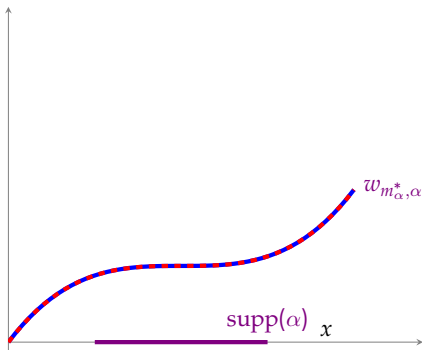
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$$w_{m,\alpha}(x) = u^{-1}(k_\alpha(x) + m).$$

Therefore,

$$w_{m,\alpha} \text{ is increasing } \forall \alpha, m \iff k_\alpha \text{ is increasing } \forall \alpha$$

Explanation: because u^{-1} is increasing.

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Proof of Claim. Recall,

$$w_{m,\alpha}(x) = u^{-1}(k_\alpha(x) + m).$$

Therefore,

$$\begin{aligned} w_{m,\alpha} \text{ is increasing } \forall \alpha, m &\iff k_\alpha \text{ is increasing } \forall \alpha \\ &\iff C \text{ is FOSD monotone.} \end{aligned}$$

Explanation: by the Fact .

Proposition. (w, α) is IC if and only if a $m \in \mathbb{R}$ exists such that

$$w(x) \leq u^{-1}(k_{\alpha}(x) + m) =: w_{m,\alpha}(x)$$

for all x , and with equality α -almost surely.

Implications:

- (i) Without loss for principal to offer $w_{m,\alpha}$ for some m .
- (ii) A cheapest contract implementing α is $w_{m_{\alpha}^*,\alpha}$ for

$$m_{\alpha}^* = -\inf k_{\alpha}(X) = 0.$$

- (iii) FOSD monotone $C \implies$ wage is increasing without loss.

EXAMPLE 2: AFFINE COSTS

Suppose we have an increasing and bounded function,

$$k : X \rightarrow \mathbb{R}_+,$$

with $k(\inf X) = 0$, and costs are given by

$$C(\alpha) = \int k(x)\alpha(dx).$$

This cost is convex and differentiable, with derivative

$$k_\alpha(\cdot) = k(\cdot).$$

EXAMPLE 2: AFFINE COSTS (CONTINUED)

In this example, (w, α) is IC if and only if an m exists s.t.

$$w(x) \leq w_{m,\alpha}(x) = u^{-1}(k(x) + m),$$

for all x , with equality α -almost surely.

EXAMPLE 2: AFFINE COSTS (CONTINUED)

In this example, (w, α) is IC if and only if an m exists s.t.

$$w(x) \leq w_{m,\alpha}(x) = u^{-1}(k(x) + m),$$

for all x , with equality α -almost surely.

Thus, cost minimizing wage for *any* α is

$$w^*(\cdot) = u^{-1} \circ k(\cdot).$$

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$$\max_{\alpha \in \Delta X} \int [x - w^*(x)] \alpha(\mathrm{d}x).$$

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- In this case, a deterministic output are optimal.

Explanation: Principal's problem is:

$$\max_{\alpha \in \Delta X} \int [x - w^*(x)] \alpha(dx).$$

So, optimal to implement a deterministic output x^* s.t.

$$x^* \in \operatorname{argmax}_{x \in X} [x - w^*(x)].$$

EXAMPLE 3: TRANSFORMATIONS OF AFFINE

Consider k as before, and let

$$\Psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$$

be increasing, convex, differentiable, and $\Psi(0) = 0$. Then,

$$C(\alpha) = \Psi \left(\int k(x) \alpha(dx) \right)$$

satisfies our assumptions. The derivative is

$$k_\alpha(\cdot) = \Psi' \left(\int k(x) \alpha(dx) \right) k(\cdot).$$

EXAMPLE 3: TRANSFORMATIONS OF AFFINE (CONT.)

In this example, (w, α) is IC if and only if an m exists s.t.

$$w(x) \leq w_{m,\alpha}(x) = u^{-1} \left(\Psi' \left(\int k(y) \alpha(dy) \right) k(x) + m \right),$$

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Given α , a cost minimizing wage is

$$w_{0,\alpha}(\cdot) = u^{-1} \left(\Psi' \left(\int k(y) \alpha(dy) \right) k(\cdot) \right).$$

Wage depends on α , so principal's problem more complex.

Profit Maximization

THE PRINCIPAL'S PROBLEM

Let $w_\alpha := w_{0,\alpha}$ be the cost minimizing wage implementing α .

The principal's problem is:

$$\max_{\alpha \in \Delta X} \left[\int x \alpha(\mathrm{d}x) - \int w_\alpha(x) \alpha(\mathrm{d}x) \right].$$

ADDITIONAL ASSUMPTIONS

Nice Agent's Payoffs. $n := u^{-1}$ is continuously differentiable.

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2nd Order Differentiability. Every α admits a continuous function $h_\alpha : X \times X \rightarrow \mathbb{R}$ such that for all β ,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [k_{\alpha + \epsilon(\beta - \alpha)}(\cdot) - k_\alpha(\cdot)] = \int h_\alpha(x, \cdot)(\beta - \alpha)(dx),$$

where convergence is in the supnorm.

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where convergence is in the supnorm.

For finite X : last two assumptions $\iff$ twice differentiability.

PRINCIPAL FIRST ORDER CONDITION

Define the function $\chi : X \rightarrow \mathbb{R}$,

$$\chi_\alpha(x) = \int h_\alpha(x, y)(n' \circ k_\alpha)(y)\alpha(dy).$$

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$$\chi_{\alpha}(x) = \int h_{\alpha}(x, y)(n' \circ k_{\alpha})(y) \alpha(dy).$$

Theorem.

A profit maximizing α^* exists. Moreover, α^* must solve

$$\max_{\alpha \in \Delta X} \int [x - w_{\alpha^*}(x) - \chi_{\alpha^*}(x)] \alpha(dx).$$

NICE PROFIT MAXIMIZING DISTRIBUTIONS

For every α , let

$$\pi_\alpha(x) := x - w_\alpha(x) - \chi_\alpha(x).$$

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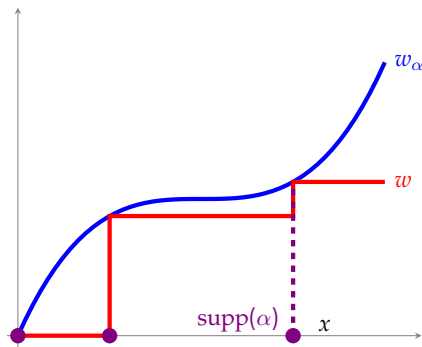
- (i) If π_α is strictly *quasiconcave* $\forall \alpha$, then $|\text{supp } \alpha^*| = 1$.
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- (iii) If $w_\alpha + \chi_\alpha$ is non-affine analytic $\forall \alpha$, $\text{supp}(\alpha^*)$ is countable.

DISCRETE DISTRIBUTIONS AND BONUSES

Remark. Suppose $X = [L, H]$, C is FOSD monotone, and $\text{supp}(\alpha)$ is countable. Then,

$$w(x) = w_\alpha \left(\max \{y \in \text{supp}(\alpha) \cup \{L\} : y \leq x\} \right)$$

is a cost minimizing contract that implements α .



EXAMPLE 3: TRANSFORMATIONS OF AFFINE (CONT.)

Consider again the cost function

$$C(\alpha) = \Psi \left(\int k(x) \alpha(dx) \right),$$

and suppose Ψ is twice continuously differentiable.

Then,

$$h_{\alpha}(x, y) = \Psi'' \left(\int k(z) \alpha(dz) \right) k(x) k(y).$$

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So,

$$\chi_{\alpha}(x) = \left[\int \Psi'' \left(\int k(z) \alpha(dz) \right) n' \left(\Psi' \left(\int k(z) \alpha(dz) \right) \right) (k(y))^2 \alpha(dy) \right] k(x).$$

EXAMPLE 3: TRANSFORMATIONS OF AFFINE (CONT.)

Hence, the Theorem implies α^* is optimal only if

$$\text{supp } \alpha^* \subseteq \operatorname{argmax}_{x \in X} [x - w_{\alpha^*}(x) - \chi_{\alpha^*}(x)],$$

where

- $w_{\alpha^*}(x) = n \left(\Psi' \left(\int k(z) \alpha^*(dz) \right) k(x) \right),$
- $\chi_{\alpha^*}(x) = [\text{positive stuff that depends on } \alpha^*] k(x).$

SPECIALIZING EXAMPLE 3

Suppose:

$$X = [0, 1], \quad u(y) = y^\rho, \quad k(x) = x^\gamma \quad \Psi(a) = \frac{a^{1+\lambda}}{1+\lambda},$$

where $\gamma, \lambda > 0$, and $\rho \in (0, 1)$.

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So for non-degenerate α , $\pi_\alpha(x) = x - w_\alpha(x) - \chi_\alpha(x)$ is

- strictly concave if $\gamma \geq 1 (> \rho)$.
- strictly convex if $\gamma \leq \rho$.
- non-constant and analytic for all parameters.

SPECIALIZING EXAMPLE 3: CONCAVE CASE

Suppose $\gamma \geq 1$, so π_α is strictly concave.

By Corollary: only deterministic distributions can be optimal.

For $x \in X = [0, 1]$, let δ_x be deterministic distribution yielding x .

Calculating $w_{\delta_x}(x) = x^{(1+\lambda)\frac{\gamma}{\rho}}$, we get x^* is optimal iff

$$\max_{x \in X} \left[x - x^{(1+\lambda)\frac{\gamma}{\rho}} \right]$$

Objective is concave, so can use FOC to get solution:

$$x^* = \left[(1 + \lambda) \frac{\gamma}{\rho} \right]^{\frac{1}{1 - (1 + \lambda) \frac{\gamma}{\rho}}}.$$

SPECIALIZING EXAMPLE 3: CONVEX CASE

Suppose $\gamma \leq \rho$, so π_α is strictly convex.

By Corollary: optimal α^* supported on $\{0, 1\}$.

For $p \in [0, 1]$, let $\alpha_p = p\delta_1 + (1 - p)\delta_0$.

In this case $w_{\alpha_p}(0) = 0$ and $w_{\alpha_p}(1) = p^{\lambda/\rho}$, so optimal p^* solves

$$\max_{p \in [0, 1]} \left[p - p^{\frac{\lambda + \rho}{\rho}} \right].$$

Again, objective is concave, so FOC gives the solution:

$$p^* = \left[\frac{\rho/\lambda}{1 + \rho/\lambda} \right]^{\rho/\lambda}.$$

Non-Smooth Costs

NON-SMOOTH COSTS

Maintain: C is convex and continuous.

Drop: C is Gateaux differentiable.

In this case, can use **directional derivative**:

$$d_{\alpha}^{+}C : \Delta X \rightarrow \mathbb{R} \cup \{\infty\},$$
$$\beta \mapsto \lim_{\epsilon \searrow 0} \frac{1}{\epsilon} [C(\alpha + \epsilon(\beta - \alpha)) - C(\alpha)].$$

(convexity of C ensures above limit is well-defined).

NON-DIFFERENTIABLE FIRST-ORDER APPROACH

Lemma*. For a bounded $v : X \rightarrow \mathbb{R}$, and $\alpha \in \mathcal{A}$,

$$\alpha \in \arg \max_{\beta \in \mathcal{A}} \left[\int v(x) \beta(\mathrm{d}x) - C(\beta) \right]$$

if and only if

$$\alpha \in \arg \max_{\beta \in \mathcal{A}} \left[\int v(x) \beta(\mathrm{d}x) - d_{\alpha}^{+} C(\beta) \right]$$

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Unfortunately, not nearly as useful without differentiability.

IMPLEMENTABILITY

Say α is **implementable** if (w, α) is IC for some $w \in W$.

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Proof.

If X is finite, C is differentiable almost everywhere.

For infinite X , show:

- Embed ΔX in $L^2(X)$.
- Show α is implementable if $\partial_{L^2} C(\alpha) \neq \emptyset$.
- Bronsted-Rockafellar: $\partial_{L^2} C(\alpha) \neq \emptyset$ holds for a dense set.

MONOTONICITY

Theorem 3. Suppose C is FOSD monotone. If (w, α) is IC, then

$$\bar{w}(x) = \sup_{y \leq x} w(y)$$

is monotone, $\bar{w} = w$ α -almost surely, and $(\bar{w}, \alpha)$ is IC.

MONTONICITY PROOF SKETCH

Suppose X is finite. Every β admits a γ_β s.t. $\beta \succeq_{FOSD} \gamma_\beta$ and

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(because (w, α) is IC)

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So, $(\bar{w}, \alpha)$ is IC.

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So, $(\bar{w}, \alpha)$ is IC. Setting $\beta = \alpha$ above gives $\bar{w} = w$ α -almost surely.

Conclusion

RELATED LITERATURE

- **Less general flexible models:** Holmstrom and Milgrom (1987), Diamond (1998), Mirrlees and Zhou (2006), Hebert (2018), Bonham (2021), Mattsson and Weibull (2022), Bonham and Riggs-Cragun (2023).
- **Flexible Monitoring:** Georgiadis and Szentes (2020), Mahzoon, Shourideh, and Zetlin-Joines (2022), Wong (2023).
- **Robust contracting:** Carroll (2015), Antic (2022), Antic and Georgiadis (2022), Carroll and Walton (2022).

FLEXIBLE MORAL HAZARD PROBLEMS

We showed that in smooth & flexible moral hazard problems:

- Parameters driving contract: k_α and u .
- Cost minimization is trivial.
- Every distribution can be implemented.
- Wages are monotone without loss.

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We showed that in smooth & flexible moral hazard problems:

- Parameters driving contract: k_α and u .
- Cost minimization is trivial.
- Every distribution can be implemented.
- Wages are monotone without loss.

Also obtained results about principal optimality.

- First order approach is valid.
- Optimality of single, binary, and discrete distributions.

Thanks!